

Programming Challenges

1. Minimum/Maximum

Write a program that asks the user to enter two numbers. The program should use the conditional operator to determine which number is the smaller and which is the larger.

2. Roman Numeral Converter

Write a program that asks the user to enter a number within the range of 1 through 10. Use a `switch` statement to display the Roman numeral version of that number.

Input Validation: Do not accept a number less than 1 or greater than 10.

3. Magic Dates

The date June 10, 1960 is special because when we write it in the following format, the month times the day equals the year.

6/10/60

Write a program that asks the user to enter a month (in numeric form), a day, and a two-digit year. The program should then determine whether the month times the day is equal to the year. If so, it should display a message saying the date is magic. Otherwise it should display a message saying the date is not magic.

4. Areas of Rectangles

The area of a rectangle is the rectangle's length times its width. Write a program that asks for the length and width of two rectangles. The program should tell the user which rectangle has the greater area, or if the areas are the same.

5. Body Mass Index

Write a program that calculates and displays a person's body mass index (BMI). The BMI is often used to determine whether a person with a sedentary lifestyle is overweight or underweight for his or her height. A person's BMI is calculated with the following formula:

$$BMI = weight \times 703 / height^2$$

where *weight* is measured in pounds and *height* is measured in inches. The program should display a message indicating whether the person has optimal weight, is underweight, or is overweight. A sedentary person's weight is considered to be optimal if his or her BMI is between 18.5 and 25. If the BMI is less than 18.5, the person is considered to be underweight. If the BMI value is greater than 25, the person is considered to be overweight.

6. Mass and Weight

Scientists measure an object's mass in kilograms and its weight in newtons. If you know the amount of mass that an object has, you can calculate its weight, in newtons, with the following formula:

$$Weight = mass \times 9.8$$

Write a program that asks the user to enter an object's mass, and then calculates and displays its weight. If the object weighs more than 1,000 newtons, display a message indicating that it is too heavy. If the object weighs less than 10 newtons, display a message indicating that the object is too light.



VideoNote
Solving the Time
Calculator
Problem

7. Time Calculator

Write a program that asks the user to enter a number of seconds.

- There are 60 seconds in a minute. If the number of seconds entered by the user is greater than or equal to 60, the program should display the number of minutes in that many seconds.
- There are 3,600 seconds in an hour. If the number of seconds entered by the user is greater than or equal to 3,600, the program should display the number of hours in that many seconds.
- There are 86,400 seconds in a day. If the number of seconds entered by the user is greater than or equal to 86,400, the program should display the number of days in that many seconds.

8. Color Mixer

The colors red, blue, and yellow are known as the primary colors because they cannot be made by mixing other colors. When you mix two primary colors, you get a secondary color, as shown here:

When you mix red and blue, you get purple.

When you mix red and yellow, you get orange.

When you mix blue and yellow, you get green.

Write a program that prompts the user to enter the names of two primary colors to mix. If the user enters anything other than “red,” “blue,” or “yellow,” the program should display an error message. Otherwise, the program should display the name of the secondary color that results by mixing two primary colors.

9. Change for a Dollar Game

Create a change-counting game that gets the user to enter the number of coins required to make exactly one dollar. The program should ask the user to enter the number of pennies, nickels, dimes, and quarters. If the total value of the coins entered is equal to one dollar, the program should congratulate the user for winning the game. Otherwise, the program should display a message indicating whether the amount entered was more than or less than one dollar.

10. Days in a Month

Write a program that asks the user to enter the month (letting the user enter an integer in the range of 1 through 12) and the year. The program should then display the number of days in that month. Use the following criteria to identify leap years:

1. Determine whether the year is divisible by 100. If it is, then it is a leap year if and only if it is divisible by 400. For example, 2000 is a leap year but 2100 is not.
2. If the year is not divisible by 100, then it is a leap year if and only if it is divisible by 4. For example, 2008 is a leap year but 2009 is not.

Here is a sample run of the program:

```
Enter a month (1-12): 2 [Enter]
Enter a year: 2008 [Enter]
29 days
```

11. Math Tutor

This is a modification of Programming Challenge 17 from Chapter 3. Write a program that can be used as a math tutor for a young student. The program should display two random numbers that are to be added, such as:

$$\begin{array}{r} 247 \\ + 129 \\ \hline \end{array}$$

The program should wait for the student to enter the answer. If the answer is correct, a message of congratulations should be printed. If the answer is incorrect, a message should be printed showing the correct answer.

12. Software Sales

A software company sells a package that retails for \$99. Quantity discounts are given according to the following table.

Quantity	Discount
10–19	20%
20–49	30%
50–99	40%
100 or more	50%

Write a program that asks for the number of units sold and computes the total cost of the purchase.

Input Validation: Make sure the number of units is greater than 0.

13. Book Club Points

Serendipity Booksellers has a book club that awards points to its customers based on the number of books purchased each month. The points are awarded as follows:

- If a customer purchases 0 books, he or she earns 0 points.
- If a customer purchases 1 book, he or she earns 5 points.
- If a customer purchases 2 books, he or she earns 15 points.
- If a customer purchases 3 books, he or she earns 30 points.
- If a customer purchases 4 or more books, he or she earns 60 points.

Write a program that asks the user to enter the number of books that he or she has purchased this month and then displays the number of points awarded.

14. Bank Charges

A bank charges \$10 per month plus the following check fees for a commercial checking account:

- \$10 each for fewer than 20 checks
- \$.08 each for 20–39 checks
- \$.06 each for 40–59 checks
- \$.04 each for 60 or more checks

The bank also charges an extra \$15 if the balance of the account falls below \$400 (before any check fees are applied). Write a program that asks for the beginning balance and the number of checks written. Compute and display the bank's service fees for the month.

Input Validation: Do not accept a negative value for the number of checks written. If a negative value is given for the beginning balance, display an urgent message indicating the account is overdrawn.

15. Shipping Charges

The Fast Freight Shipping Company charges the following rates:

Weight of Package (in Kilograms)	Rate per 500 Miles Shipped
2 kg or less	\$1.10
Over 2 kg but not more than 6 kg	\$2.20
Over 6 kg but not more than 10 kg	\$3.70
Over 10 kg but not more than 20 kg	\$4.80

Write a program that asks for the weight of the package and the distance it is to be shipped, and then displays the charges.

Input Validation: Do not accept values of 0 or less for the weight of the package. Do not accept weights of more than 20 kg (this is the maximum weight the company will ship). Do not accept distances of less than 10 miles or more than 3,000 miles. These are the company's minimum and maximum shipping distances.

16. Running the Race

Write a program that asks for the names of three runners and the time it took each of them to finish a race. The program should display who came in first, second, and third place.

Input Validation: Only accept positive numbers for the times.

17. Personal Best

Write a program that asks for the name of a pole vaulter and the dates and vault heights (in meters) of the athlete's three best vaults. It should then report, in order of height (best first), the date on which each vault was made and its height.

Input Validation: Only accept values between 2.0 and 5.0 for the heights.

18. Fat Gram Calculator

Write a program that asks for the number of calories and fat grams in a food. The program should display the percentage of calories that come from fat. If the calories from fat are less than 30% of the total calories of the food, it should also display a message indicating that the food is low in fat.

One gram of fat has 9 calories, so

Calories from fat = fat grams * 9

The percentage of calories from fat can be calculated as

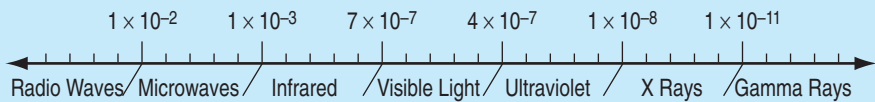
Calories from fat ÷ total calories

Input Validation: Make sure the number of calories and fat grams are not less than 0. Also, the number of calories from fat cannot be greater than the total number of calories. If that happens, display an error message indicating that either the calories or fat grams were incorrectly entered.

19. Spectral Analysis

If a scientist knows the wavelength of an electromagnetic wave, he or she can determine what type of radiation it is. Write a program that asks for the wavelength of an electromagnetic wave in meters and then displays what that wave is according to the

chart below. (For example, a wave with a wavelength of 1E-10 meters would be an X-ray.)



20. The Speed of Sound

The following table shows the approximate speed of sound in air, water, and steel.

Medium	Speed
Air	1,100 feet per second
Water	4,900 feet per second
Steel	16,400 feet per second

Write a program that displays a menu allowing the user to select air, water, or steel. After the user has made a selection, he or she should be asked to enter the distance a sound wave will travel in the selected medium. The program will then display the amount of time it will take. (Round the answer to four decimal places.)

Input Validation: Check that the user has selected one of the available choices from the menu. Do not accept distances less than 0.

21. The Speed of Sound in Gases

When sound travels through a gas, its speed depends primarily on the density of the medium. The less dense the medium, the faster the speed will be. The following table shows the approximate speed of sound at 0 degrees centigrade, measured in meters per second, when traveling through carbon dioxide, air, helium, and hydrogen.

Medium	Speed (Meters per Second)
Carbon Dioxide	258.0
Air	331.5
Helium	972.0
Hydrogen	1,270.0

Write a program that displays a menu allowing the user to select one of these four gases. After a selection has been made, the user should enter the number of seconds it took for the sound to travel in this medium from its source to the location at which it was detected. The program should then report how far away (in meters) the source of the sound was from the detection location.

Input Validation: Check that the user has selected one of the available menu choices. Do not accept times less than 0 seconds or more than 30 seconds.

22. Freezing and Boiling Points

The following table lists the freezing and boiling points of several substances. Write a program that asks the user to enter a temperature and then shows all the substances that will freeze at that temperature and all that will boil at that temperature. For example, if the user enters -20 the program should report that water will freeze and oxygen will boil at that temperature.

Substance	Freezing Point (°F)	Boiling Point (°F)
Ethyl alcohol	−173	172
Mercury	−38	676
Oxygen	−362	−306
Water	32	212

23. Geometry Calculator

Write a program that displays the following menu:

```
Geometry Calculator
```

1. Calculate the Area of a Circle
2. Calculate the Area of a Rectangle
3. Calculate the Area of a Triangle
4. Quit

```
Enter your choice (1-4):
```

If the user enters 1, the program should ask for the radius of the circle and then display its area. Use the following formula:

$$\text{area} = \pi r^2$$

Use 3.14159 for π and the radius of the circle for r . If the user enters 2, the program should ask for the length and width of the rectangle and then display the rectangle's area. Use the following formula:

$$\text{area} = \text{length} * \text{width}$$

If the user enters 3 the program should ask for the length of the triangle's base and its height, and then display its area. Use the following formula:

$$\text{area} = \text{base} * \text{height} * .5$$

If the user enters 4, the program should end.

Input Validation: Display an error message if the user enters a number outside the range of 1 through 4 when selecting an item from the menu. Do not accept negative values for the circle's radius, the rectangle's length or width, or the triangle's base or height.

24. Long-Distance Calls

A long-distance carrier charges the following rates for telephone calls:

Starting Time of Call	Rate per Minute
00:00–06:59	0.05
07:00–19:00	0.45
19:01–23:59	0.20

Write a program that asks for the starting time and the number of minutes of the call, and displays the charges. The program should ask for the time to be entered as a floating-point number in the form HH.MM. For example, 07:00 hours will be entered as 07.00, and 16:28 hours will be entered as 16.28.

Input Validation: The program should not accept times that are greater than 23:59. Also, no number whose last two digits are greater than 59 should be accepted. Hint: Assuming num is a floating-point variable, the following expression will give you its fractional part:

```
num - static_cast<int>(num)
```

25. Mobile Service Provider

A mobile phone service provider has three different subscription packages for its customers:

Package A: For \$39.99 per month 450 minutes are provided. Additional minutes are \$0.45 per minute.

Package B: For \$59.99 per month 900 minutes are provided. Additional minutes are \$0.40 per minute.

Package C: For \$69.99 per month unlimited minutes provided.

Write a program that calculates a customer's monthly bill. It should ask which package the customer has purchased and how many minutes were used. It should then display the total amount due.

Input Validation: Be sure the user only selects package A, B, or C.

26. Mobile Service Provider, Part 2

Modify the Program in Programming Challenge 25 so that it also displays how much money Package A customers would save if they purchased packages B or C, and how much money Package B customers would save if they purchased Package C. If there would be no savings, no message should be printed.

27. Mobile Service Provider, Part 3

Months with 30 days have 720 hours, and months with 31 days have 744 hours. February, with 28 days, has 672 hours. You can calculate the number of minutes in a month by multiplying its number of hours by 60. Enhance the input validation of the Mobile Service Provider program by asking the user for the month (by name), and validating that the number of minutes entered is not more than the maximum for the entire month. Here is a table of the months, their days, and number of hours in each.

Month	Days	Hours
January	31	744
February	28	672
March	31	744
April	30	720
May	31	744
June	30	720
July	31	744
August	31	744
September	30	720
October	31	744
November	30	720
December	31	744